Mapping (Version 1.0)
Matter public key cryptography SHALL be based on Elliptic Curve Cryptography (ECC) with the elliptic curve: `secp256r1` defined in Section 2.4.2 of SEC 2. (This curve is also referred to as `NIST P-256` or `prime256v1` in FIPS 186-4 and NIST 800-186.)
`PrivateKey` is an opaque data type to hold either the private key or any handle or reference that allows other primitives to access the corresponding private key.
`PublicKey` is an opaque data type to hold the public key or any handle or reference that allows other primitives to access the corresponding public key. A public key is a point on the elliptic curve. (At places in the specification where public keys are to be explicitly transmitted, the format in which they are transmitted is specified.)
`int CRYPTO_GROUP_SIZE_BITS := 256`
`int CRYPTO_GROUP_SIZE_BYTES := 32`
`int CRYPTO_PUBLIC_KEY_SIZE_BYTES : = (2 * CRYPTO_GROUP_SIZE_BYTES) + 1 = 65` is the size in bytes of the public key representation when encoded using the uncompressed public key format as specified in section 2.3 of SEC 1.
`struct { PublicKey publicKey; PrivateKey privateKey; } KeyPair;`

3.5.2. Key generation

`Crypto_GenerateKeyPair()` is the function to generate a key pair.

Function and description

```
KeyPair
Crypto_GenerateKeyPair()
```

Generates a key pair and returns a `KeyPair`.

Mapping (Version 1.0)

```
Crypto_GenerateKeypair() :=
    KeyPair ECCGenerateKeypair()
```

`ECCGenerateKeypair()` SHALL generate a key pair according to Section 3.2.1 of SEC 1.

3.5.3. Signature and verification

`Crypto_Sign()` is used to sign a message, and `Crypto_Verify()` is used to verify a signature on a message.

